Martin Dang

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Education

University of California San Diego Expected June 2027
B.S. in Math-Computer Science

Work Experience

EQTY Lab April 2026 – Present
Engineering Intern

- Built a prompt-injection scanner system for AI email agents: a dependency-free Python scanner (**9 obfuscation signals**, recursive multi-layer decoding) feeding a **3-stage ML cascade**.
- Built a verifiable-compute layer connecting the EQTY SDK to **NVIDIA FLARE 2.8** federated learning: every training, update, and aggregation step is cryptographically signed into an offline-verifiable proof manifest.
- Built the foundation of two open-source agent skills (Claude Code/Codex) for AI supply-chain auditability:
  - **eqty-instrument**: patches Python repos (incl. **LangChain**, **LangGraph**, **DeepAgents**) using the EQTY SDK to emit signed run records, guided by an agent-generated control-flow graph.
  - **eqty-manifest**: teaches the agent to read and parse a manifest so it can understand a whole run from the manifest alone, and cryptographically verifies those records, including signatures, content hashes, and hardware attestation.

Stanford Farm Games September 2024 – January 2026
App Developer

- Sole developer of Farm Games, a **React Native/TypeScript** iOS/Android app on **Firebase** for Stanford undergraduates, replacing manual attendance and point tracking.
- Includes real-time leaderboard, multi-day event calendar, team rosters, QR check-ins with transactional points, and FCM reminders, secured by role-based auth and emulator-tested Firestore rules.
- Secured backend with Firebase Auth (user/admin roles), server-enforced Firestore/Storage rules validated by an emulator test suite, App Check, and forced-update gating.

Projects

Warden | *TypeScript, Chrome MV3 (WXT), Cloudflare Workers/R2, Ed25519*

- Open-source Chrome extension blocking phishing via a staged cascade — allowlist → **declarativeNetRequest** blocklist + Safe Browsing → credential-gated heuristics — plus a popup/redirect hijack guard.
- Built the feed backend as an hourly-cron **Cloudflare Worker** publishing versioned snapshots and deltas behind an **Ed25519**-signed manifest with per-file **SHA-256** checksums, giving one root of trust against list-tampering.

SHA256_pwcracker | *C, OpenSSL*

- Built a **SHA-256** dictionary attack tool in C with **OpenSSL** (EVP API), cracking hashes against the rockyou wordlist with overflow-safe mixed-radix enumeration of case and leet-speak variants.
- Hardened against malformed input (strict hex/length validation, CRLF-safe parsing) and validated correctness with **14 unit tests** covering edge cases like empty strings, length boundaries, and reverse-leet substitutions.

HTTP Chat Server | *C*

- Built a modular HTTP chat server in C — split across HTTP parsing, request handling, and core logic — supporting post/react/edit/history endpoints over plain GET.
- Enforced input validation (field length limits, 400 error handling) and scaled to **100,000+ stored messages** with per-message reaction tracking.

CTF Practice

- **OverTheWire: Bandit Wargames** — Completed **30+ levels** covering Linux fundamentals, permissions, and basic exploitation via CLI.
- **picoCTF** — Solved cryptography, reverse engineering, and web exploitation challenges in a CTF developed by Carnegie Mellon, applying core cybersecurity concepts.
- **TryHackMe** — Browser-based, gamified online platform for learning and building cybersecurity skills through hands-on labs and real-world attack simulations.

Skills

Languages: Swift, Java, C, Python, SQL, Bash
Tools: WireShark, XCode, Firebase, OpenSSL, Linux (CLI)
Skills: Web Exploitation, Prompt Injection, Cryptography, Reverse Engineering, Forensics, Binary Exploitation